

GRAVITYSORT · REQUIREMENTS

GravitySort requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Status from GVS-CAL-001 v0.4 for the constructable design (GVS-DDR-003); R11 not met (95.7 kg, every load under 30 kg); R9 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R9 target \$450; R12 restated for the pedal drive; R11 met with the lighter frame tube; status from GVS-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from GVS-CAL-001; R7 pack range aligned with MotionCore and GVS-DDR-001 item 6; MotionCore cost \$335; proposed changes to R11 and R12 listed, awaiting Amish
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

GravitySort requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. At TRL 3 each has been checked by calculation or design review in GVS-CAL-001 v0.4, for the constructable design of GVS-DDR-003 (every part can be made and fixed to the next): one is not met (R11, mass), three are at risk (R2, R5, R6), two cannot be verified before testing (R4, R14) and seven are met on paper or by design review. R9 is reported against the value-engineering target: the constructable design is USD 103 over it. Recovery can only be verified by testing with real or spiked ore, which is TRL 4 work and on hold by Amish's instruction.

Table 1. Requirements. Status is from GVS-CAL-001 (docs/04-calcs/results.csv).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (GVS-CAL-001)
R1	Use no mercury	No amalgamation step, no mercury-coated plates or traps, no step that needs mercury to recover gold from the concentrate	Design review of the flowsheet	Met (design review)
R2	Process a small group's daily ore	200 kg/h or more of ore milled to below 2 mm, as slurry of 25 to 35 % solids, for 8 h (1.6 t per day)	Bowl capacity and cycle calculation	At risk: 1.55 t per shift after three 5 min flush stops; 1.6 t needs 206 kg/h
R3	Hold fine gold in the bowl	Bowl acceleration adjustable from 40 to 80 G at the riffle rings (about 600 to 850 rpm at 100 mm radius), with speed shown to the operator	Drive ratio and speed calculation	Met on paper: 598 to 846 rpm at a 50 to 71 rpm cadence; speed display (BOM item 19)
R4	Recover fine gold	Of free gold in the screened feed: 80 % or more of 75 to 1,000 µm gold and 50 % or more of 38 to 75 µm gold in the bowl concentrate; 60 % or more of all gold in the ore to smelted gold overall	Test with spiked or characterized ore (TRL 4)	Not verifiable at TRL 3; at risk. Settling check passes; bed behaviour needs testing
R5	Make a concentrate that can be smelted without mercury	Bowl mass pull 0.5 % or less of feed; shaking table reduces one day's bowl concentrate (about 5 kg) to 100 g or less	Mass balance calculation; later test	At risk: bowl pull 0.33 % is met; 5.3 kg per day needs 53:1 on the table, likely two passes
R6	Run on pedal power	Full throughput at 60 W or less at the pedals, at a cadence of 55 to 70 rpm	Power estimate from slurry, fluidization water and bearing losses	At risk: 48.2 W at 60 G is met; 62.8 W at 80 G is not; union seal drag assumed
R7	Run on a small motor	Full throughput from the MotionCore 250 W reference drive on any 20 to 58 V pack that MotionCore accepts, with at least 3 times power margin	Power estimate	Met on paper: 5.2 times at 60 G, 4.0 times at 80 G; 0.55 kWh per shift
R8	Use little water	1.5 m ³ /h or less at 200 kg/h, and tolerate recirculated water with fine silt	Water balance calculation	Met on paper: 1.19 m ³ /h; a 3/4 in fluidization hose is needed (GVS-CAL-001 section 6)
R9	Keep parts cost near the value-engineering target	Value-engineering target \$455 for parts, excluding the MotionCore module and battery: a hypothetical control target, not a limit (Amish, 2026-10-01; was \$350, then \$450; GVS-DDR-002)	Priced BOM	Estimated cost of the constructable design \$558: over the value-engineering target by USD 103
R10	Be built in a local workshop	Welding, drilling and hand tools only; no lathe; bowl liner cast in a printed mold	Design review of every part	Met (design review): set-screw bearing inserts, a taper bush and a welded nipple on a tube spindle avoid the lathe; liner cast on a six-segment printed core (GVS-DDR-003); casting route unproven

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (GVS-CAL-001)
R11	Travel to site	Breaks into loads of 30 kg or less, carried by two people; total 80 kg or less; assembled with hand tools in 30 min or less	Mass estimate from the model	Not met: 95.7 kg in six loads, 16 kg over 80 kg, for the constructable design (GVS-DDR-003); every load is under 30 kg (heaviest 25.2 kg); assembly time not verified
R12	Guard every moving part	Bowl covered by a lid guard during running; belts, chains and the table head fully guarded; with the motor, bowl speed limited to 900 rpm or less by the drive ratio and the MotionCore speed limit; with the pedals, which gearing cannot cap, a burst safety factor of 10 or more for the bowl, jacket and rings at the highest reachable speed (1,200 rpm at a 100 rpm cadence) and the bowl speed shown to the rider; bowl stops within 15 s of stopping the drive (restated per GVS-DDR-002)	Design review, burst calculation and safety checklist	Met on paper: guards, lid interlock and a disc brake (0.6 s stop; about 20 s coasting); motor capped by its 1.2:1 step-up and the MotionCore limit; lowest pedal-case safety factor 17 at 1,200 rpm; speed display item 19. Containment of a liner fragment not verified
R13	Clean up quickly and securely	Bowl concentrate flushed into a lockable container in 5 min or less without tools; table concentrate drops into a lockable tray	Design review; later timed trial	Met on paper: toolless lid clamps, lid lifts clear of the feed pipe; lockable concentrate box under the table's far end; time not verified
R14	Last in abrasive service	Bowl liner and riffles replaceable in 30 min; liner life 500 h or more (estimate to be checked)	Wear data for cast polyurethane; later test	Not verifiable at TRL 3

■ Assumptions

- Ore is milled to below 2 mm by the user's existing mill and screened on the GravitySort hopper screen.
- Free, liberated gold in the 38 to 1,000 µm range. Gold locked in sulfides or finer than about 20 µm is lost to tailings by any gravity method, so R4 applies to free gold only.
- Reference ore for the daily balance: 5 g/t, 1.6 t per day. Real grades vary widely (about 1 to 20 g/t).
- Healthy adult pedalling at 60 W or less for sessions of about 1 h, with operators taking turns.
- Budget: the \$455 in `project.yaml` is a value-engineering target, a hypothetical control target and not a limit (Amish, 2026-10-01) (raised from \$350 to \$450 by Amish's decision of 2026-09-25 and to \$455 on 2026-09-26, GVS-DDR-002) covers GravitySort parts. The MotionCore kit and reference motor (\$335: \$265 kit plus \$70 motor, per MTC-CAL-001) and a battery are shared lab components, budgeted with MotionCore.
- The pack for the motor option is left to the user (GVS-DDR-001 item 6).

■ Changes decided on 2026-09-25

Amish accepted the recommendations on 2026-09-25 (GVS-DDR-002):

- **R9:** target raised from \$350 to \$450 (item 1).
- **R12:** for the pedal drive, "limited to 900 rpm or less by gearing" is replaced by a burst safety factor of 10 or more at the highest reachable speed (1,200 rpm at a 100 rpm cadence) plus the speed display (item 9).
- **R11:** target unchanged at 80 kg; the frame, pedal outrigger and table stand move to 25 x 25 x 1.5 mm tube and the mass is re-estimated at 77.3 kg (item 10).

■ Change decided on 2026-09-26

- **R9:** target raised from \$450 to \$455 to cover the priced BOM (budget approved by Amish, GVS-DDR-002). Not met to met on paper.

■ Still proposed, awaiting Amish

These are tracked in the design decisions register, GVS-DEC-001 (docs/06-design-decisions.md).

- **R2:** restate the target as 200 kg/h of feed time, or raise the design feed to about 210 kg/h (GVS-DDR-001 item 13). No option was recommended.
- **R11:** restate the total as 100 kg or less, keeping loads of 30 kg or less, or look for about 16 kg of savings (GVS-DDR-003, A1). Recommendation: restate the total.